Process Control

Pre-Lab

Unit Operations Lab #5

10/29/2019

Group #3, Tuesday PM

Daniela Estarita

Huda Usman

Nisha Patel

Mike Topalis

Introduction

1. Experimental Objective

The objective of this lab is to understand the fundamental principles of the process control. The Armfield PCT50 was used to study the fluid-based process to demonstrate single PID loop using the Level as the measured variable. In this experiment our goal is to study the effect of changes made to the system by applying the disturbances to the process to analyze the appropriate controller settings.

Exercise A:

PCT50 Level Process will be utilized to control the water level in the process. Water is manually altered by changing the speed of the pump from which water is introduced into the system. This exercise will be utilized to analyze the characteristics of the process levels containing a range of in and out flow to and from the process vessel.

Exercise C:

Proportional controller will be utilized to control the water level using an automatic level control by changing the speed of the pump. This exercise will help us determine the level process using P and P+I only control to change the speed of the pump.

2. Experimental Detail

Serial Number	Apparatus	Explanation
1.	Armfield PCT50 Level Control	Level Control Process which uses water as the working fluid.
2.	Lower Sump Tank	Water is stored
3.	Upper process vessel	Water is transferred here to perform the level controls
4.	Speed centrifugal pump	It is used to pump the water from lower sump tank to the upper process vessel
4.	Inline Ball Valves (CV1, CV2, CV3)	CV1 - allows to vary the water flow rate coming in the process valve CV2, CV3 - allows to vary the flow rate constantly to meet a fixed level
5.	Electronic pressure sensor	It measures the pressure in the process valve relative to the atmospheric pressure
6.	Solenoid valve (SOL)	It is used to start and stop flow through the second outlet
7.	Interchangeable Orifice	It allows to stabilize the flow of both valve outlets (CV2, CV3) at a fixed value

Figures of Apparatus

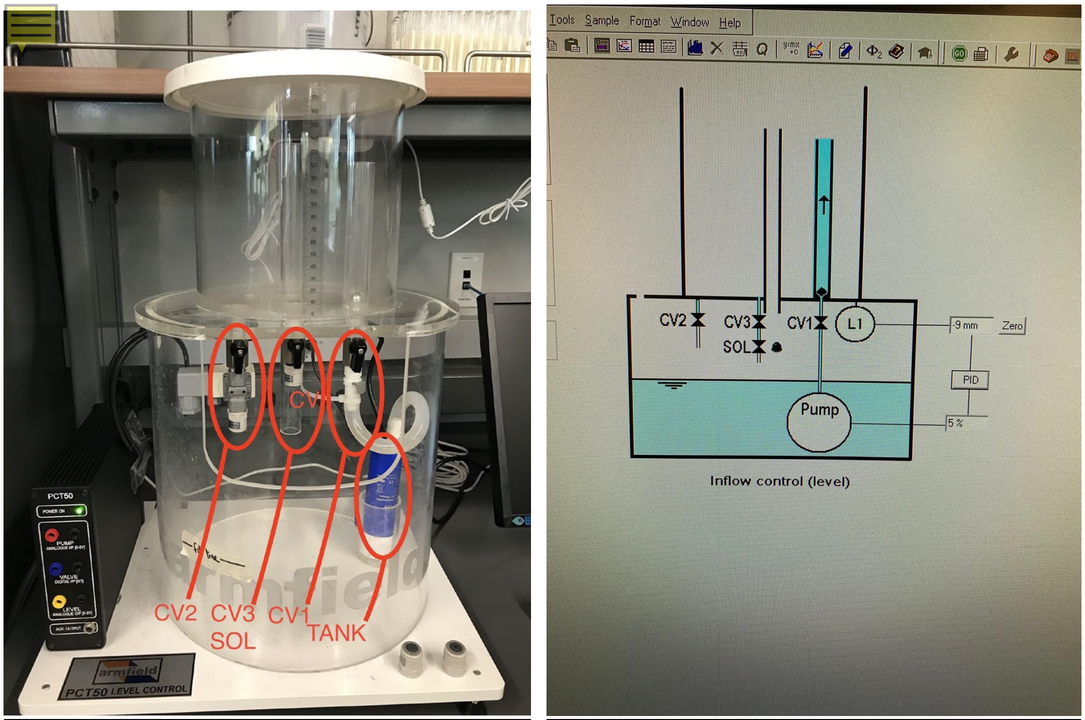

Figure 1: PCT50 Level Control Process

Figure 2: Full Unit (w/ Armfield Software Computer)

2.2 Materials and Supplies

Serial Number	Materials and Supplies	Explanation
1.	Water	Working fluid used to study level control experiment
2.	Bucket	Used to dump water inside Armfield.

2.3 Experimental Plan

EXERCISE A:

1. Increase the speed of the pump and observe when the water flow exceeds the out flow.
2. When the pump speed is matched with the out flow the level should be kept constant.
3. Choose the "GO" icon to login in the data with the level constant in the process.
4. Open the solenoid valve. Observe the level fall due to the outflow from the solenoid valve.
5. When the level goes down to 20 mm, note the pump speed then elevate the speed of the pump until the original level is restored.
6. Close the solenoid valve.
7. When the level goes at or above 20 mm note the speed of the pump and then reduce the speed of the pump until the original level is restored.
8. Choose the "STOP" icon to finish data entry and graph the results.
9. Observe the water level going up and down when solenoid is open and closed and what steps were made to restore the level.
10. Reduce the speed of the pump and drain the vessel.
11. Create new sheet and start data login.
12. Now, increase the speed of the pump until the level rises upto 150 mm and allow vessel to fill.
13. When the vessel is empty, use 75 % the speed of the pump and fill the vessel. Stop the pump once the pump reaches 150 mm and drain the vessel.
14. Click the "STOP" icon to finish data login
15. Generate graphs using Armfield software and observe the effects of different pump speeds on the rates of filling the vessel.
16. Same experiment can be done using different orifice fittings having different sizes.

Exercise C:
Experiment Setup:

1. Assure the computer and connecting tubes are working/connected correctly according to the installation reference.
2. The sump pump must be filed with clean DI water.
3. Power supply must be connected to 24 V IN
4. Initially fully open outlet valve (CV2) to fit a 4 mm diameter orifice below and a 2 mm orifice below the solenoid valve
5. USB must be connected to the computer with PCT50 software to then select "Experiment Ex1: Inflow Control".
6. Check the Virtual COM is properly enabled from the inlet valve (CV1).
7. Click View -> Power On to obtain the water level from the vessel, and then zero it.
8. Begin setting the pump speed to 50% to let the water flow to the vessel.
9. Be aware that as the level of water rises the inlet valve (CV1) must be adjusted accordingly to allow a steady water level among the vessel.
10. Flow of CV1 must be the same as CV2.
11. Once completed drain the vessel via CV2 valve by setting pump speed 0%.

P Controller Procedure:

1. Go to PID-> set Proportional Band P to 200%, Integral Time I to 0, Derivation time to 0 and Set PointL 75 mm - > Apply.
2. Click "Automatic Mode of Operation"-> Go
3. Open solenoid valve by clicking on "mimic" button
4. An offset will occur as the water level rises in which the P value will be adjusted accordingly (as directed in the next steps). Let the water level settle close to the solenoid valve before proceeding.

12. The proportional control via the computer will be used to control the steady level of water in the vessel by automatically varying the speed of inflow from the pump:
    -set set point to 100 mm -> "Apply"
13. Let the water level settle then set point to 75 mm by the same process in 12.

14. Plots of different Proportional values will be created to observe offsets from the water level target.
15. From offset points the stability of "P" will be modified accordingly to better obtain stable values and minimize off set points after initially adjusting P to 100% by "PID" - > Apply.
16. Close the solenoid valve once the offset reduces slightly.
17. Once a steady offset is obtained repeat steps above for each direction:
    -set P to 50%
    -set P to 20%
    -set P to 10%

-set P to 5%
-set P to 1%

18. Click “Stop” -> “View” to obtain graph of results.
19. Click “Format”-> “Plot Level L1” -> “Set point” on 1 axis -> “Controller Output” on 2 axis
20. Observe a setting that will allow a stable control and reduce offsets then apply.
21. If time allows repeat above procedure with different orifice sizes to CV2 and solenoid valves.

P+I Controller Procedure:

1. Create new results sheet.
2. Select PID box, set the Proportional Band to value selected previously, integral time I to 0 seconds and Derivative time to D to 0.
3. Set point to 75mm and click Apply.
4. Begin the data logging
5. Select the automatic mode for the operation and allow water to stabilize.
6. Change the I term to 100 in PID controller and click apply.
7. After the level stabilized to the set point, open the solenoid valve to add disturbance.
8. Once again let the level settle and then close the solenoid and allow to settle.
9. Change the set point to 100mm and observe.
10. Once the level settles, again set the point back to 75mm.
11. Change the I term to 50 seconds and observe.
12. Once the level settles, open and then close the solenoid valve as mentioned above, to add disturbances in the process.
13. Once again adjust the set point to 100mm and observe the level. Again let the level settle and then return the value of set point to 75 mm.
14. Continue reducing the Integral time and inducing disturbance and set point changes till the level becomes unstable.

References:

[1] Mccabe W.L., Smith J.C., Harriot P. *Unit Operations in Chemical Engineering*. 7th Edition pp. 46-52, (2005)
[2] UIC, Unknown. *ChE 381: Unit Operations Laboratory Process Control*. University of Illinois at Chicago. Che 381 Unit Operations I under Professor Zdunek, Alan

Project Safety Evaluation

Project Title: Process Control

Date: October 29, 2019

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Be cautious of pouring DI in main tank by the computer. Do not let the pump speed overflow the water in tank that one cannot stop in time to avoid spills onto the computer and electrical sensors.
Potential mechanical hazards?	No	If yes, identify conditions:
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	yes	If yes, specify type: Nitrile

MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Water
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	25	Room temperature
Pressure (atm)	1	Pressure is room pressure
Water Level (mm)	30 & 150	Water level must not exceed 30 mm below the “normal use” cut offline initially. Water level not exceed 150 mm when pump is at 75% speed.
Pump Speed (%)	75	Be cautious to stop water level from overflowing at the high speed.
Safety Controls	*yes*	*If yes, provide description: The pressure regulator must stay clamped on the wall for support*

		Unscrewing the plastic fitting to change the orifice and head in the process vessel. Must be tighten fully by hand.
Valves/latches	Yes	When priming the pump, the drain tank must be fully closed before filling. Only release latch connector when temporarily disconnecting the tube from the system. *Note the solenoid valve is the main controlling valve for the process.

I have read relevant background material for the Unit Operations Laboratory entitled: "Viscosity of Liquids" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

signature of team member & date

Daniela Estarita 10/29/19

Michael Topalis 10/29/19

Huda Usman 10/29/19

Nisha Patel 10/29/19